Michael Simon

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Education

- **University of California, Los Angeles** — Los Angeles, CA
  - *Ph.D. Student in Computer Science* — *Sept. 2026 – Present*
  - *M.S. in Computer Science* — *Sept. 2024 – June 2026*
  - *B.S. in Computer Science* — *Oct. 2020 – Mar. 2024*
  - Coursework: Advanced Deep Learning, Natural Language Processing, Neurosymbolic Reasoning, Human-Computer Interaction

Experience

- **Computation and Discovery Lab, UCLA** — Los Angeles, CA
  - *Graduate Student Researcher* — *Feb. 2025 – Present*
  - Designed Syrup, a concept-role-graph intermediate representation for bidirectional human-AI scientific reasoning; co-first-authored the resulting paper
  - Built the accompanying interactive system and evaluated it through qualitative and quantitative user studies with researchers and PhD students
- **Excavaite Inc** — Palo Alto, CA
  - *CTO & Co-founder* — *Feb. 2024 – May 2026*
  - Built a terabyte-scale patent corpus with RaBitQ-indexed vector search on Postgres and VectorChord, powering retrieval for agentic workflows
  - Designed agentic systems for patent leakage prevention, patent drafting, and claim chart generation; built automated evals to gate model and prompt changes
  - Secured a six-figure Fortune 500 pilot and raised pre-seed funding
  - Hired and managed an 8-person engineering team, owning architecture and technical direction end to end
- **VAIL Lab, UCLA** — Los Angeles, CA
  - *Undergraduate Researcher* — *June 2023 – June 2024*
  - Researched diffusion models for synthetic driving scene generation, resulting in SimGen (NeurIPS 2024)
  - Created a unified terabyte-scale driving dataset by combining 4K YouTube footage with open-source datasets
- **Confluent** — Mountain View, CA
  - *Software Engineer Intern, Kafka Metadata Team* — *June 2023 – Aug. 2023*
  - Built a last-resort Java metadata recovery tool, cutting Kafka cluster recovery time from hours to minutes
- **Meta** — Menlo Park, CA
  - *Software Engineer Intern, Metrics Platform Governance* — *June 2022 – Aug. 2022*
  - Created a centralized Hack (PHP) code-generation framework, cutting data onboarding time from ∼1 hr to ∼5 min; parallelized generation for a further 4x speedup
- **Agrofocal Technologies** — San Jose, CA
  - *Machine Learning Intern (part-time)* — *June 2021 – Nov. 2023*
  - Implemented a mixed-precision quantization pipeline, increasing inference throughput by over 80% and halving latency
  - Implemented a state-of-the-art PyTorch architecture, improving object detection accuracy by over 0.15 mAP
  - Removed the need for a discrete GPU on edge devices via a hardware-acceleration SDK, saving $1,000s per device

Selected Publications

- **Towards Bidirectional Human–AI Scientific Reasoning: From Concept-Role Graphs to Negotiated Interpretation**. *M.S. thesis, UCLA, June 2026.*
- **SimGen: Simulator-conditioned Driving Scene Generation**. *NeurIPS 2024* (second author).